

Analytical Modeling of Wind Turbine Dynamics Using Symbolic Lagrangian Mechanics

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ABSTRACT

The size and flexibility of modern offshore wind turbines lead to new challenges in terms of aeroelastic phenomena threatening structural integrity. This thesis hopes to establish a series of analytical wind turbine models of increasing complexity to study the fundamental trends that governing how system frequencies and aerodynamic damping vary with rotor rotational speed and ambient wind velocity. The research aims to find the physical reasons behind non-intuitive dynamic behavior, such as why the second tower mode varies significantly depending on whether the rotor is modeled as stiff or flexible, and how gyroscopic effects influence the frequencies of the turbine's yawing and tilting modes.

The equations of motion for the multi-body system are derived using the energy-based Lagrange method. For more complex models, the Python package `sympy` is used as a symbolic framework to systematically generate, manipulate, and linearize these equations. By performing an eigenvalue analysis on the resulting linear systems, the frequency and damping characteristics can be determined. The analytical and numerical outputs of these reduced-order models will be compared against more advanced, high-fidelity numerical tools, providing a transparent and validated understanding of wind turbine aeroelasticity.

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I. INTRODUCTION

A. Motivation and the Shift to Offshore Wind

The global transition toward sustainable energy systems has placed wind energy among the leading technologies for power generation. To meet growing energy demands and maximize efficiency, the industry has shifted begun shifting its focus from land-based systems to offshore environments. This transition is driven by their physical advantages; specifically, offshore wind turbines offer an opportunity to move sustainable energy generation to locations where higher energy outputs are possible, as "Wind velocity is higher and more dependable at offshore locations" and "offshore wind energy [being] characterized by higher power density" [1]. To effectively harness this energy, offshore wind turbines are larger in size and complexity compared to their onshore counterparts. Understanding their dynamic behavior of larger turbines becomes increasingly important for ensuring structural integrity and operational efficiency. This literature review aims to establish a foundation for using Lagrangian mechanics to derive equations of motion (EOMs) investigate the aeroelastic stability of wind turbines and how they vary with rotational and wind speeds.

This drive for offshore deployment is underscored by fundamental physical principles. The available power in the wind, P , increases dramatically with the wind velocity, v :

$$P \propto v^3 \quad (1)$$

Furthermore, the power captured by a turbine is directly proportional to the swept area of its rotor, which scales with the square of the blade length, r :

$$A \propto r^2 \quad (2)$$

As a result, small increases in wind speed or blade length yield large gains in power output. Modern offshore turbines feature extraordinarily long blades and towering support structures to maximize energy capture; however, this unprecedented geometric upscaling introduces an engineering challenge in modeling their dynamic behavior. Today's multi-megawatt turbines no longer behave as rigid bodies, rather operating as highly flexible structures. This increased flexibility introduces complex aeroelastic phenomena — interactions between aerodynamic forces and structural motion — not present in rigid systems. Ensuring their stability of these structures against phenomena such as classical flutter and stall-induced vibrations, is a primary design constraint [2].

B. Problem Statement: Higher-Order Modes and Analytical Challenges

While the first-order modes of wind turbines (such as the first fore-aft tower bending or first flapwise blade bending) are well-documented and standard in design load cases, higher-order modes remain an important area of study which can impact turbine stability. As turbines become larger for offshore purposes, they also lose rigidity,

causing modes like the second tower mode to gain prominence in influencing the dynamic response of the structure. The literature identifies that these higher-order modes can exhibit very low or even negative aerodynamic damping under specific operational conditions, leading to potential aeroelastic instabilities such as stall-induced vibrations or classical flutter [2].

The damping mechanisms for the second tower mode are often non-intuitive — defying “common sense” or classic newtonian expectations. They mechanisms arise from complex aeroelastic couplings with the rotor that vary non-linearly with rotational speed and pitch settings [2]. The continuous rotation of the massive turbine rotor introduces gyroscopic couplings fundamentally altering the system’s dynamic response, heavily influencing the frequencies of the turbine’s yawing and tilting modes and causing frequency splitting [3].

Currently, the wind energy industry relies heavily on sophisticated, high-fidelity numerical simulators (such as OpenFAST) to analyze these modes. While these tools are highly accurate, these tools often obscure the underlying mathematical relationships governing the system’s behavior. They function essentially as numerical black boxes, with parameters going in and results coming out. This makes it difficult to identify the root physical causes of instability when a structural frequency jumps or damping goes negative during a simulation.

C. Research Objectives and Methodological Approach

The primary objective of this thesis is to systematically establish reduced-order, analytical wind turbine models of increasing complexity to uncover the fundamental physical trends governing aeroelastic stability. Rather than relying solely on black-box simulations, this research aims to provide explicit, mathematically transparent explanations for non-intuitive phenomena, such as why the second tower mode varies so significantly depending on whether the rotor is modeled as stiff or flexible, and how gyroscopic effects dynamically dictate the frequencies of the yawing and tilting modes.

To accurately describe the motion of these complex systems, this work relies on Lagrangian mechanics. This energy-based methodology utilizes the principle of least action to analyze complex, rotating reference frames much more efficiently than the basic $F = ma$ approach used in Newtonian mechanics [4]. To bridge the gap between continuous flexible structures and discrete mathematics, this project employs the Rayleigh-Ritz method to map “the infinite number of DOF necessary to describe a flexible body onto a reduced and converging set of shape functions” [5].

Because deriving equations of motion manually becomes intractable for flexible, rotating systems, the literature emphasizes the utility of symbolic computational frameworks over purely numerical solvers. The Python package `sympy` is utilized as a symbolic framework to systematically generate and manipulate these equations, allowing for the creation of “glass-box” models where the underlying physical relationships remain perfectly transparent [6]. By performing an eigenvalue analysis on these resulting linear systems, the frequency and damping characteristics

can be mapped against varying rotor speeds (Ω) and wind velocities (V). Ultimately, the analytical and numerical outputs of these transparent, symbolic models will be validated against advanced numerical simulations.

II. WIND TURBINE MODELING

A. Structural Dynamics and Symbolic Lagrangian Mechanics

The modeling of rotating flexible systems, such as wind turbines, requires a mechanical framework capable of handling complex, interconnected kinematic chains. While Newtonian mechanics ($F = ma$) provides a straightforward approach to analyzing the motion of simple systems, it becomes exceedingly complex when applied to multibody systems featuring rotating reference frames and varying constraint forces. Instead, the Lagrangian approach is vastly preferred for these applications [4].

Lagrangian mechanics relies on an energy-based formulation, utilizing the kinetic energy ($\mathcal{T}$) and potential energy ($\mathcal{V}$) to bypass the need to explicitly calculate internal constraint forces at every joint. For a wind turbine, the total kinetic energy of any rigid component (such as the nacelle or a rigid rotor) is the sum of its translational and rotational kinetic energies:

$$\mathcal{T} = \frac{1}{2}m(\vec{v} \cdot \vec{v}) + \frac{1}{2}\vec{\omega}^T \mathbf{J} \vec{\omega} \quad (3)$$

where m is the mass, $\vec{v}$ is the absolute velocity of the center of mass, $\vec{\omega}$ is the absolute angular velocity vector, and $\mathbf{J}$ is the inertia tensor [4].

Potential energy ($\mathcal{V}$) is typically derived from gravitational effects, elastic deformation, and aerodynamic forces. For example, the potential energy of a rigid body in a gravitational field is given by:

$$\mathcal{V} = mgh \quad (4)$$

where g is the acceleration due to gravity and h is the height of the center of mass above a reference level.

By formulating the Lagrangian ($\mathcal{L} = \mathcal{T} - \mathcal{V}$), the system's equations of motion can be found using the Euler-Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_{i,nc} \quad (5)$$

Here, q_i represents the set of generalized coordinates (the degrees of freedom), and $Q_{i,nc}$ represents the generalized non-conservative forces, such as aerodynamic loads and structural damping [4].

Within the `sympy.physics.mechanics` environment used in this thesis, these differential equations can be rearranged into a more easily manageable matrix form:

$$\mathbf{M}(\mathbf{q}, t) \ddot{\mathbf{q}} = \mathbf{F}(\mathbf{q}, \dot{\mathbf{q}}, t) \quad (6)$$

This form with the mass matrix ($\mathbf{M}$) and the forcing vector ($\mathbf{F}$), has advantages in computational efficiency and clarity when analyzing a system's dynamics. This formulation serves as the necessary foundation for the subsequent linearization and eigenvalue analysis of the dynamic system.

B. Modeling Approaches

Modern wind turbine blades and towers bend and twist, so they cannot be modeled simply as rigid bodies. However, treating them as continuous flexible structures results in highly complex Partial Differential Equations (PDEs) with an infinite number of degrees of freedom. An important part of modeling wind turbine dynamics is therefore reducing the system into a mathematically manageable form. This can be achieved through the Rayleigh-Ritz method.

This method works by separating where the blade bends from when it bends. It maps the infinite number of degrees of freedom onto a reduced set of known spatial shape functions ($\phi(x)$) multiplied by time-dependent generalized coordinates ($q(t)$) [5]. For a mass element dm at a distance x along an undeformed blade, its position in the local frame subjected to deflection is approximated as:

$$\mathbf{r}_{dm}(x, t) \approx x\hat{\mathbf{r}}_z + \phi(x)q(t)\hat{\mathbf{r}}_x \quad (7)$$

By integrating the velocity and strain of these differential elements over the length of the flexible body, the continuous system is discretized into a finite set of Ordinary Differential Equations (ODEs). The blade's structural properties reduce to generalized mass (GM_b) and stiffness (K_b) that plug into the Lagrangian energy equations.

Crucially, this Assumed Modes method forms the exact theoretical backbone of industry-standard aeroelastic simulators, such as the ElastoDyn module within the NREL OpenFAST framework. However, while OpenFAST computes these integrals numerically behind the scenes, deriving them using symbolic frameworks allows for the generation of explicit mass and stiffness matrices [6]. This symbolic approach removes approximation errors that happen during derivation and provides mathematical expressions to analyze, showing how structural elasticity couples with rotational kinematics.

C. Aeroelastic Stability, Damping, and Whirling Modes

As wind turbine structures become increasingly flexible, they become susceptible to a range of aeroelastic instabilities that can severely compromise structural integrity. Hansen categorizes these instability mechanisms into distinct phenomena, identifying stall-induced vibrations and classical flutter as the two most critical risks for modern turbines [2].

Classical flutter is an instability driven by the coupling of flapwise bending and torsional twisting, primarily influenced by the blade tip speed, torsional stiffness, and the chordwise position of the center of gravity [2].

Conversely, stall-induced vibrations are driven by variations in the aerodynamic forces. While blade modes typically benefit from high aerodynamic damping due to their large surface area interacting with the airflow, tower modes present a unique challenge. In conditions where the turbine enters aerodynamic stall, the effective damping can become negative. Without sufficient structural damping to counteract this, self-excited vibrations grow in amplitude until structural failure occurs [2].

Analyzing the stability of these higher-order modes requires an understanding of how they interact with the rotating rotor. When a flexible tower bends, it does not oscillate in isolation. The continuous rotation of the massive rotor creates gyroscopic couplings that split the tower's natural frequencies, a phenomenon known as "whirling." This interaction results in both a forward whirl (where the natural frequency increases with rotor speed) and a backward whirl (where the frequency decreases) [3].

This frequency splitting is highly sensitive to the rotational speed (Ω) of the turbine. Capturing these non-linear dependencies accurately is essential for predicting the response of the structure. The symbolic derivation and linearization of the EOMs developed in this thesis will allow a clear, analytical mapping of whirling frequencies and damping characteristics.

D. Shape Functions and Finite Element Methods

To model the continuous deformation of wind turbine blades and towers, structural dynamics uses the concept of shape functions. A shape function is a continuous mathematical expression that maps the discrete displacements and rotations of specific points—known as nodes—to any continuous point within the body of a structure. This is relevant to wind turbines because the beam cross section is not uniform along the length of the blade. As a result, the deformation and rotation of the blade does not occur in a simple, uniform manner. Shape functions capture the bending of more complicated structures.

On top of that, because a flexible wind turbine blade is a continuous medium, it theoretically possesses an infinite number of degrees of freedom. Shape functions allow engineers to discretize this continuous medium, approximating the true, infinite-degree-of-freedom deformation field using a finite, manageable set of variables.

The primary benefit of utilizing shape functions lies in their ability to bridge the gap between physical reality and computational limitations. By defining the spatial deformation of a component through these functions, the spatial variables can be separated from the time-dependent generalized coordinates. This separation allows the spatial integrals required by Lagrangian mechanics to be evaluated analytically before a simulation even begins. Consequently, this method transforms complex partial differential equations (PDEs) into a set of ordinary differential equations (ODEs). This reduction in computational overhead allows aeroelastic simulators to run dynamic analyses of highly flexible, multi-megawatt turbines in reasonable timeframes without sacrificing accuracy.

The foundation of shape functions for beams is derived from the Euler-Bernoulli beam theory, which require shape functions to maintain continuity for physical displacement and rotation across the beam. For a 1D beam with two nodes, there are four boundary conditions — the displacement and rotation at each node. As such, the displacement of the beam is modeled using a four-term cubic polynomial, known as cubic Hermite shape functions, with four unknown coefficients (a_0, a_1, a_2, a_3):

$$w(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

To find these coefficients, the polynomial is evaluated at the boundaries of the beam, where the displacement and rotation are defined. These boundary conditions occur at the two nodes of the beam, located at $x = 0$ and $x = L$:

$$w(0) = w_1$$

$$\frac{dw}{dx}(0) = \theta_1$$

$$w(L) = w_2$$

$$\frac{dw}{dx}(L) = \theta_2$$

Where w_1 and w_2 are the displacements at the first and second nodes, respectively, and θ_1 and θ_2 are the rotations at those nodes. Solving this system of equations for the coefficients in terms of these boundary conditions yields the shape functions that can be used to interpolate the displacement along the beam. Factoring the resulting polynomial by the boundary conditions gives the cubic Hermite shape functions. Defining the non-dimensional coordinate $\xi = x/L$ — normalizing the beam so that it always runs from 0 to 1 regardless of its actual length — the shape functions for the beam element emerge to isolate the unit translations and unit rotations at the nodes. The unit rotation shape functions interpolate pure physical rotation while keeping the displacement at the nodes zero:

$$N_1(\xi) = 1 - 3\xi^2 + 2\xi^3$$

$$N_3(\xi) = 3\xi^2 - 2\xi^3$$

The unit rotation shape functions do the opposite, interpolating the pure physical rotation of the beam while restricting the displacement at the nodes to zero:

$$N_2(\xi) = L(\xi - 2\xi^2 + \xi^3)$$

$$N_4(\xi) = L(-\xi^2 + \xi^3)$$

These shape functions can then be used to express the displacement of any point along the beam as a function of

the nodal displacements and rotations, allowing for the accurate modeling of the beam's deformation under various loading conditions showing the change in displacement and rotations along the length of the beam and through time. The resulting expression for the displacement of the beam at any point x along its length is given by:

$$w(x) = N_1(\xi)w_1 + N_2(\xi)\theta_1 + N_3(\xi)w_2 + N_4(\xi)\theta_2$$

where w_1 , w_2 , θ_1 , and θ_2 are the time-dependent nodal displacements and rotations. This is more visible when the shape functions are expressed in the form:

$$w(x, t) = \sum_{i=1}^4 N_i(x)q_i(t) = \mathbf{N}(x)\mathbf{q}(t)$$

$$\mathbf{q}(t) = [w_1(t), \theta_1(t), w_2(t), \theta_2(t)]^T$$

This representation of the beam's deformation allows for the evaluation of the kinetic and potential energy required by Lagrangian mechanics, which in turn can be used to derive the EOMs for the system. Integrating these shape functions into the Finite Element Method (FEM) improves the modeling of wind turbines compared simpler methods. While a basic approach tries to guess one continuous mathematical curve for an entire tower or blade, FEM divides the structure into many smaller, interconnected beam elements, each governed by these cubic shape functions. This localized approach allows the model to account for the variations in mass, chord length, and structural stiffness that occur from the root of a blade to its tip. By evaluating the Lagrangian energy equations element by element using these unit translation and rotation functions, engineers can assemble highly accurate global mass and stiffness matrices. Integration FEM shape functions provides the mathematical rigor required to predict aeroelastic phenomena, such as flutter and whirling, in modern offshore wind energy systems.

III. PROJECT EVALUATION AND MANAGEMENT

A. Evaluation

This Honors Thesis will be evaluated based on the completion of both a Research Manuscript and functional computational models. Specifically, Professor Branlard will assess the work based on measurable goals including the symbolic derivation of Equations of Motion for wind turbine models of increasing complexity, and the analytical validation of the frequency and damping trends within these models. Progress will be assessed weekly through review of derived equations and code snippets.

B. Communication and Workflow

To maintain consistent progress, a regular meeting schedule is maintained with the faculty sponsor. Meetings are held weekly or bi-weekly, focusing on resolving mathematical bottlenecks, reviewing python model logic, and

setting goals for the upcoming week. Between these scheduled reviews, approximately ten hours per week are dedicated to research, coding, and writing to maintain the necessary pace for completion.

C. Project Timeline

From January through March, I will continue work on the Honors Thesis, ensuring that research results are reviewed continuously during weekly meetings. The formal writing process will proceed iteratively, aiming to create a draft of the Research Manuscript by Early May for Professor Branlard to review. All the while I intend to refine the computational models and finalize the written manuscript. The properly formatted Research Manuscript along with the associated Python repository, will be submitted via CHC Paths by the end of classes.

IV. METHODOLOGY

A. Symbolic Derivation Pipeline

The equations of motion for the wind turbine models are generated through a symbolic derivation pipeline built on the Python `sympy.physics.mechanics` module. The workflow begins by defining the system's reference frames, point locations, and generalized coordinates ($q(t)$), after which the kinetic energy ($\mathcal{T}$) and potential energy ($\mathcal{V}$) are constructed symbolically. The pipeline then evaluates the Euler-Lagrange equations,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_{i,nc}, \quad (8)$$

automatically executing the required partial differentiation and algebraic simplification—steps that become quite difficult to complete by hand for systems involving complex three-dimensional rotational kinematics and flexible bodies. The acceleration terms ($\ddot{q}$) are then extracted to recast the nonlinear ordinary differential equations into standard matrix form:

$$M(q, t)\ddot{q} = F(q, \dot{q}, t). \quad (9)$$

Because the symbolic relationships between mass, geometry, and rotation are preserved exactly throughout this process, the resulting model is fully transparent to inspection and analysis.

Listing 1. Simplified SymPy Derivation Pipeline

B. Linearization Procedure

Although the nonlinear matrices M and F accurately capture the system's large-deflection dynamics, frequency and stability analysis requires linearization about a steady operating point. The equilibrium state (q_0) is defined by

a constant nacelle tilt (θ_0), a constant rotor speed ($\dot{\psi} = \Omega$), and zero initial blade deflection ($q_0 = 0$). A small perturbation (δq) is then superimposed on this equilibrium,

$$q(t) = q_0 + \delta q(t), \quad (10)$$

and substituting this perturbed state into the nonlinear equations and retaining only first-order terms yields the linearized equations of motion:

$$M\delta\ddot{q} + C\delta\dot{q} + K\delta q = 0. \quad (11)$$

Here, the mass matrix (M) represents the inertial properties of the system, and the stiffness matrix (K) encodes both structural elasticity and centrifugal and gravitational stiffening effects. The damping matrix (C) collects structural and aerodynamic damping contributions alongside the skew-symmetric gyroscopic terms arising from rotor rotation. The interplay among these matrices—in particular, the negative aerodynamic damping that develops during stall—governs the onset of aeroelastic instability.

C. Eigenvalue Analysis

To extract natural frequencies and modal damping characteristics from the linearized system, the second-order matrix equation is recast as a first-order state-space system,

$$\dot{x} = Ax, \quad (12)$$

where the state vector $x = [\delta q, \delta \dot{q}]^T$ and the state matrix is

$$A = \begin{bmatrix} 0 & I \\ -M^{-1}K & -M^{-1}C \end{bmatrix}. \quad (13)$$

The eigenvalues (λ) of A take the complex form

$$\lambda = \sigma \pm i\omega_d, \quad (14)$$

where the imaginary part (ω_d) is the damped natural frequency and the real part (σ) is the modal decay rate. The damping ratio follows as

$$\zeta = \frac{-\sigma}{\sqrt{\sigma^2 + \omega_d^2}}. \quad (15)$$

This eigenvalue analysis is embedded in a parametric sweep over rotor speed (Ω) and wind velocity (V), and the resulting frequencies and damping ratios are plotted to expose phenomena such as frequency veering and negative damping. The analytical predictions are then benchmarked against eigenvalues computed by high-fidelity OpenFAST (ElastoDyn/AeroDyn) simulations at identical operating conditions to validate the model.

V. ANALYSIS AND RESULTS

To achieve the objectives of this thesis, I will be using the sympy library in Python to perform the symbolic derivation of the equations of motion. This process will begin with a familiarization of the library with progressively more complex systems. A systematic approach is used to ensure the accuracy of the derivations, starting with simple systems and building up to more complicated models that incorporate flexibility and rotation. This methodical approach allows for the validation of the framework at each step.

The equations of motion for five models were generated and validated against benchmarks. These models include the Simple Pendulum, a chaotic Double Pendulum, a 2-DOF Single Blade Pendulum on a Cart, a Fixed Nacelle with a Flexible Rotating Blade, and a Flexible Tower with a Rotating Rigid Blade. Following validation, the pipeline that was developed while deriving these models was applied to derive two additional models. These models were the Tilting Nacelle with a Rigid Rotating Blade and the Tilting Nacelle with a Flexible Rotating Blade. The following sections provide a detailed description of the derivation process for each of these models.

A. Tilting Nacelle, Rigid Blade

1) *Model Description and Degrees of Freedom:* The tilting nacelle model captures the structural dynamics of a tower-top assembly with two degrees of freedom: a fore-aft tilt angle θ representing rotation in the x-z plane of the nacelle about the pivot point O , and an azimuth angle ψ representing rotation of the rigid blade relative to the nacelle. A schematic of the model, including coordinate frames and mass locations, is shown in Fig. 1.

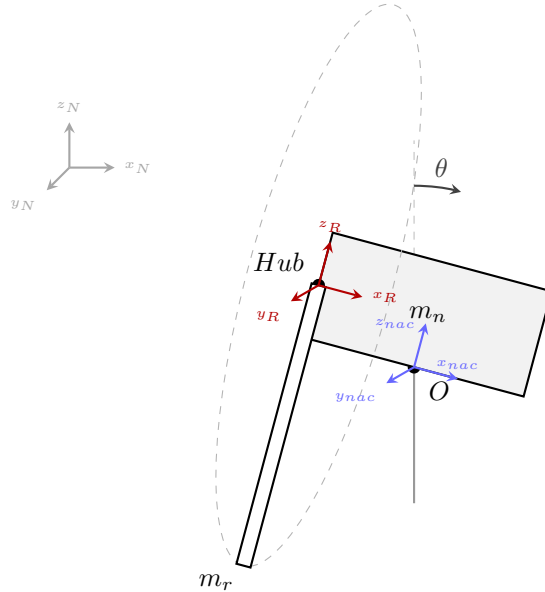


Fig. 1. Schematic of the tilting nacelle model showing coordinate frames, pivot point O , and mass locations.

The nacelle center of mass is located at offsets (X_n, Z_n) from the pivot, the hub at (X_h, Z_h) , and the rotor center of mass at radius Z_r from the hub. The inertial parameters used throughout this analysis are listed in Table I.

TABLE I
MODEL 1 INERTIAL AND GEOMETRIC PARAMETERS.

Parameter	Symbol	Value	Units
Rotor mass	m_b	<value>	kg
Nacelle mass	m_n	<value>	kg
Nacelle yaw inertia	J_{yyN}	<value>	kg m ²
Rotor inertia (xx)	J_{xxR}	<value>	kg m ²
Rotor inertia (yy)	J_{yyR}	<value>	kg m ²
Rotor inertia (zz)	J_{zzR}	<value>	kg m ²
Hub offset (x)	X_h	<value>	m
Hub offset (z)	Z_h	<value>	m

2) *Linearized Equations of Motion:* The nonlinear mass matrix $M(q, t)$ and forcing vector $F(q, \dot{q}, t)$ derived symbolically in Section V-A are linearized about the equilibrium state $(\theta_0, \dot{\psi} = \Omega, \delta q = 0)$ to yield the constant-coefficient system

$$M_0 \delta \ddot{q} + C_0 \delta \dot{q} + K_0 \delta q = 0, \quad (16)$$

where $q = [\theta, \psi]^T$. The three coefficient matrices evaluated at the operating point are:

$$M_0 = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}, \quad C_0 = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}, \quad K_0 = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix}. \quad (17)$$

Several features of these matrices warrant physical interpretation. The off-diagonal entries of C_0 are skew-symmetric—that is, $C_{12} = -C_{21}$ —and scale linearly with Ω ; these are the gyroscopic coupling terms generated by the spinning rotor, which transfer energy between the tilt and azimuth modes. In the stiffness matrix K_0 , the diagonal entry K_{11} contains both the structural restoring stiffness k_t and a gravitational term proportional to $g \cos \theta_0$; the latter increases the effective stiffness in the upright position and represents gravitational stiffening of the tilt mode. A centrifugal stiffening contribution proportional to Ω^2 also appears in K_{11} , causing the tilt natural frequency to rise with rotor speed as discussed in Section V-A3.

3) *Frequency Analysis:* The natural frequencies of the linearized system are obtained by computing the eigenvalues of the state matrix A over a swept range of rotor speeds Ω , as described in Section ?? . Fig. 2 shows the resulting frequency diagram.

Fig. 2. Natural frequencies of the tilt mode (θ) and azimuth mode (ψ) as a function of rotor speed Ω for Model 1. Forward and backward whirl branches are labeled.

At zero rotor speed, the two modes are degenerate and share a common natural frequency set by the structural stiffness and system inertia. As Ω increases, gyroscopic coupling breaks this degeneracy and the two branches

diverge: the forward whirl branch—in which the tilt precession is co-directional with rotor spin—rises in frequency, while the backward whirl branch falls. This frequency splitting is a classical consequence of the gyroscopic terms in C_0 and grows in proportion to Ω . Simultaneously, the centrifugal stiffening term in K_{11} causes a gradual upward shift in the mean tilt frequency, which is visible as an upward bias of both branches relative to their zero-speed values.

4) *Damping Analysis:*]

Fig. 3 shows the damping ratio ζ of the tilt mode as a function of wind speed V at a representative rotor speed.

Fig. 3. Damping ratio ζ of the tilt mode as a function of wind speed V for Model 1. The horizontal dashed line marks $\zeta = 0$.

At low wind speeds, aerodynamic damping augments the structural damping and the tilt mode remains well-damped. As wind speed increases toward stall, the slope of the lift curve $\partial C_L / \partial \alpha$ diminishes and eventually becomes negative (see Section ??). Because aerodynamic damping is proportional to this slope, the total damping in C_0 decreases monotonically with wind speed beyond this point. The model predicts that ζ crosses zero at approximately $V = \langle \text{value} \rangle$ m/s (indicated in Fig. 3), beyond which the tilt mode is aeroelastically unstable. This onset is consistent with the negative-damping mechanism described in Section ??: once the rotor operates in the stall regime, perturbations in tilt angle produce aerodynamic forces that amplify rather than oppose the motion.

5) *Key Findings from Model 1:* The tilting nacelle model demonstrates that gyroscopic coupling from the spinning rotor is the dominant mechanism driving frequency splitting between the forward and backward whirl branches, and that centrifugal stiffening produces a measurable increase in the tilt natural frequency with rotor speed. On the damping side, the model clearly identifies a critical wind speed beyond which aerodynamic stall drives the tilt mode unstable, establishing the physical link between rotor aerodynamics and structural instability. However, because the blade is treated as a rigid body, this model cannot account for the in-plane and out-of-plane flexibility modes of the blade or their coupling with the nacelle dynamics; capturing those effects—and the additional damping and frequency shifts they introduce—requires the flexible blade model presented in the following section.

B. *Tilting Nacelle, Flexible Blade*

Combining the rigid 3D nacelle tilt dynamics with the approximation of a flexible blade establishes a 3-DOF aeroelastic skeleton. This architecture, consisting of nacelle pitch (θ), rotor azimuth (ψ), and flapwise blade bending (q), is required to capture the massive coupling terms inducing aerodynamic instabilities.

Due to the complex interaction between the pitching nacelle plane and the spinning, deflecting blade, the mathematical derivation of the kinetic energy (T) generates extensive velocity cross-terms. By utilizing symbolic coefficient extraction during the derivation process, these cross-terms can be mapped cleanly back onto the blade's

pre-calculated modal integrals. To maintain document readability, the generalized mass matrix (M) and forcing vector (F) are presented below using modal shorthand:

$$M = \begin{bmatrix} GM_b & 0 & 0 \\ 0 & J_{yyb} \cos^2(\psi) + J_{yy n} + J_{zzb} \sin^2(\psi) + m_b \left(X_h^2 + Z_h^2 + 2Z_h(Z_b \cos(\psi) + q \sin(\psi)) + (Z_b \cos(\psi) + q \sin(\psi))^2 \right) + m_n (X_n^2 + Z_n^2) & X_h m_b (-Z_b \sin(\psi) + q \cos(\psi)) \\ 0 & X_h m_b (-Z_b \sin(\psi) + q \cos(\psi)) & J_{xxb} + m_b (Z_b^2 + q^2) \end{bmatrix} \quad (18)$$

[illegible]

VI. CONCLUSION

The review of current literature shows a clear distinction in the modeling of wind turbine dynamics. On one hand, numerical simulators provide comprehensive data but often obscure the driving physical mechanisms behind complex instabilities. On the other hand, symbolic models have emerged as powerful tools for isolating specific dynamic behaviors [6]. These symbolic frameworks are identified as particularly valuable for understanding the fundamental trends of aeroelastic instabilities, such as flutter and stall-induced vibrations, which pose significant risks to the structural integrity of modern, flexible offshore turbines [2].

Despite the proven utility of these symbolic methods, a distinct gap remains in their application to higher-order structural modes. While the fundamental blade and tower modes are well-documented in stability analyses, the second tower mode has received less analytical attention. However, existing studies often rely on simulation to observe these effects, rather than deriving them from first principles to expose the underlying mathematical dependencies.

This thesis hopes to bridge this gap by establishing a theoretical foundation for the symbolic derivation of wind turbine equations of motion. These equations will subsequently be linearized to perform eigenvalue analysis, allowing for the precise mapping of frequency and damping trends as functions of rotational speed and wind velocity. Ultimately, this work aims to contribute to the safer design of next-generation offshore turbines by demystifying the aeroelastic behavior of their increasingly flexible structures.

APPENDIX

A. Simple Pendulum

The initial step will be deriving the equations of motion of a simple pendulum using Lagrangian mechanics. This will serve as a concept for the symbolic derivation process and allow for the validation of the method against known results. I will derive the equations for both a point mass suspended at the end of the pendulum and a rigid body pendulum.

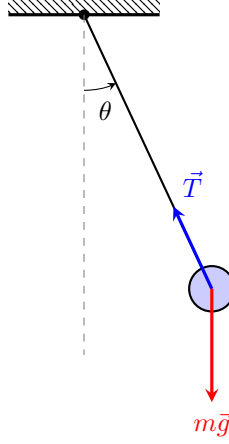


Fig. 5. Free Body Diagram of a simple pendulum with a point mass. The generalized coordinate θ is defined relative to the vertical, and the two primary forces are the tension $\vec{T}$ and gravity $m\vec{g}$, which contains a radial ($mg \cos \theta$) and tangential ($mg \sin \theta$) component.

Point Mass:

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0 \quad (20)$$

Rigid Body: For a uniform rod of mass m and length l , $I = \frac{1}{3}ml^2$. The resulting EOM is:

$$\ddot{\theta} + \frac{3g}{2l} \sin \theta = 0 \quad (21)$$

B. Double Pendulum

The same processes were applied to a double pendulum, two pendulums attached end to end. This system is another well known system, making it another good proof of concept for the symbolic derivation process. The equations of motion for the double pendulum are more complex than the simple pendulum, as they involve the interaction between the two masses and their respective angles. The angles are θ_1 and θ_2 calculated relative to the previous angle.

Rigid Body EOM:

$$\begin{bmatrix} \frac{1}{3}l_1^2m_1 + l_1^2m_2 + l_1l_2m_2 \cos \theta_2 + \frac{1}{3}l_2^2m_2 & l_2m_2 \left(\frac{1}{2}l_1 \cos \theta_2 + \frac{1}{3}l_2 \right) \\ l_2m_2 \left(\frac{1}{2}l_1 \cos \theta_2 + \frac{1}{3}l_2 \right) & \frac{1}{3}l_2^2m_2 \end{bmatrix} \ddot{\mathbf{q}} = \mathbf{F} \quad (22)$$

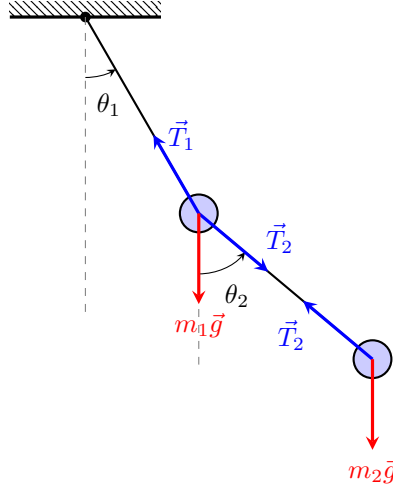


Fig. 6. Free Body Diagram of a double pendulum. The coordinates θ_1 and θ_2 are defined relative to the vertical. Mass m_1 is acted upon by gravity $m_1\vec{g}$ and the tensions from both rods, while m_2 is acted upon by $m_2\vec{g}$ and the tension $\vec{T}_2$.

C. 2-DOF Rigid Nacelle-Rotor Model

A single blade model, working on increasing complexity, was derived using a 2-DOF "pendulum on a cart" analogy. This model consists of a translational degree of freedom x representing the tower's fore-aft motion, and a rotational degree of freedom ψ representing the azimuth of the blade. The mass is separated into a nacelle mass (m_n), and a rotor mass ($m_s + m_b$).

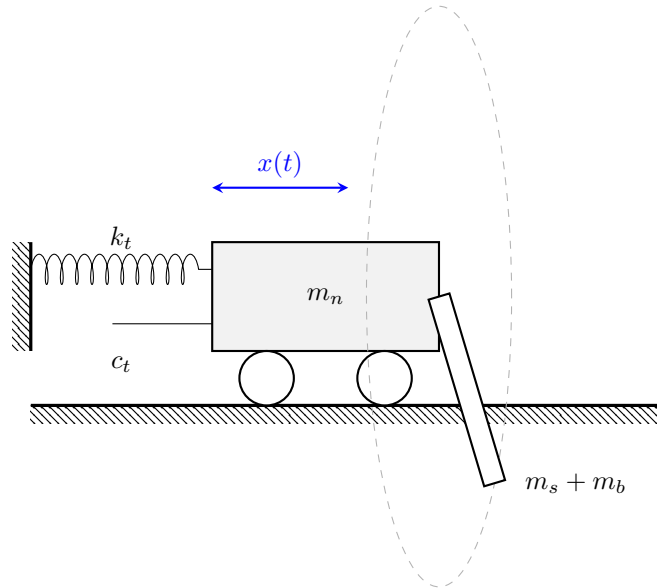


Fig. 7. Refined 2-DOF model with azimuthal rotation (ψ).

$$\begin{bmatrix} m_b + m_n + m_s & 0 \\ 0 & I_{rxx} + (Y_r^2 + Z_r^2)(m_b + m_s) \end{bmatrix} \ddot{q} = \begin{bmatrix} -c_t \dot{x} - k_t x \\ -g(m_b + m_s)(Y_r \cos \psi - Z_r \sin \psi) \end{bmatrix} \quad (23)$$

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D. Fixed Nacelle, Flexible Blade

A flexible blade rotating about a fixed nacelle model was developed as a simplified model for flexible blade analysis. This model employs a flexing degree of freedom (q) to represent the deflection of the blade, and a rotational degree of freedom (ψ) representing the azimuth. The nacelle is assumed to be rigid and stationary for this derivation to isolate the rotation and blade bending.

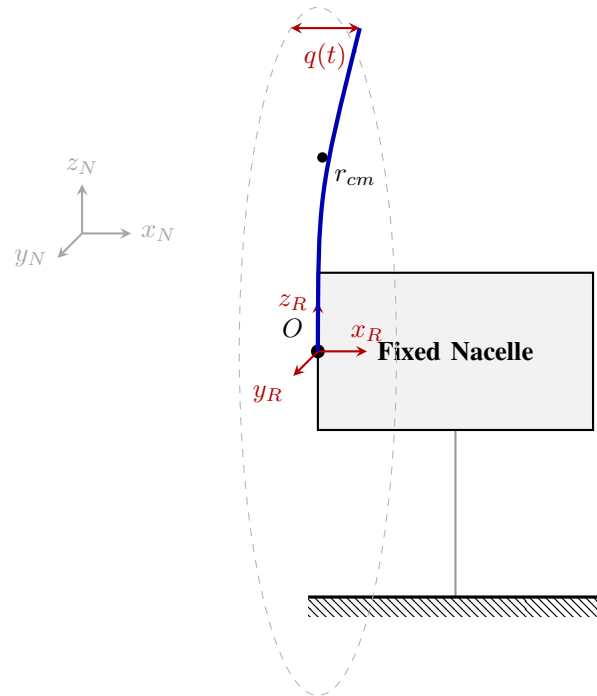


Fig. 8. Side-on perspective of the rotating flexible blade.

The system’s kinetic energy consists of rotational inertia for the hub and blade (J_{hub}, J_b) and flexural energy associated with the generalized modal mass (GM_b). Potential energy accounts for the modal elastic restoration of

the blade (K_b) and gravitational effects on the blade's center of mass (r_{cm}). The mass matrix M and forcing vector F cleanly map these dependencies:

$$M = \begin{bmatrix} GM_b & 0 \\ 0 & J_b + J_{hub} \end{bmatrix} \quad (24)$$

$$F = \begin{bmatrix} -K_b q(t) \\ gm_b r_{cm} \sin \psi(t) \end{bmatrix} \quad (25)$$

This model serves as the basis for calculating the blade's natural frequencies and analyzing how the gravitational load fluctuates as a function of the azimuth ψ .

E. Flexible Tower, Rigid Blade

To increase complexity the next model incorporates tower flexibility. The tower's bending is modeled as a single generalized coordinate (q_{T1}). This transition from a simple rigid tilt to a flexible representation involves the use of modal parameters, including the modal mass (GM_{T11}), modal stiffness (K_{eT11}), and modal damping (D_{eT11}).

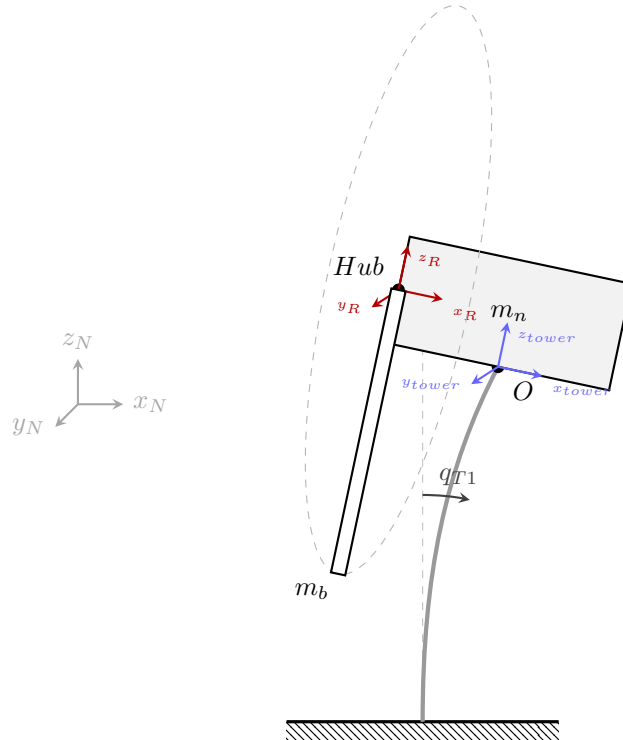


Fig. 9. Schematic of the Flexible Tower model with modal deflection q_{T1} .

To introduce structural coupling and enable the analysis of whirling phenomena, this model expands to include a flexible tower supporting a rotating blade. The tower's first fore-aft bending mode is modeled using the generalized

coordinate q_{T1} . This model incorporates the tower's generalized modal mass (GM_{T11}), modal stiffness (k_t), and modal damping coefficient (d_t). The rotor remains a rigid body with an inertial frame R rotating at an azimuth ψ .

The mass matrix M and forcing vector F accurately capture the trigonometric coupling terms (e.g., $X_h Z_b m_b \sin \psi$) generated by the non-linear interaction of the tower's lateral deflection and the rotor's dynamic azimuth position:

The equations of motion were derived by considering the kinetic energy of the flexible tower, the translating and rotating nacelle, and the rotating rotor. The mass matrix M and forcing vector F are expressed in their full symbolic form below.

$$M = \begin{bmatrix} GM_{T11} + J_{yyN} + J_{yyb} \cos^2 \psi + J_{zzb} \sin^2 \psi + m_b(X_h^2 + Z_b^2 \cos^2 \psi + 2Z_b Z_h \cos \psi + Z_h^2) + m_n(X_n^2 + Z_n^2) & X_h Z_b m_b \sin \psi \\ X_h Z_b m_b \sin \psi & J_{xxb} + Z_b^2 m_b \end{bmatrix} \quad (26)$$

$$F = \begin{bmatrix} J_{yyb} \sin(2\psi) \dot{\psi} \dot{q}_{T1} - J_{zzb} \sin(2\psi) \dot{\psi} \dot{q}_{T1} + Z_b m_b (-X_h \cos \psi \dot{\psi} + Z_b \sin(2\psi) \dot{q}_{T1} + 2Z_h \sin \psi \dot{q}_{T1}) \dot{\psi} - d_t \dot{q}_{T1} + g m_b (X_h \cos q_{T1} + Z_b \sin q_{T1} \cos \psi + Z_h \sin q_{T1}) + g m_n (X_n \cos q_{T1} + Z_n \sin q_{T1}) - k_t q_{T1} \\ (-J_{yyb} \cos \psi \dot{q}_{T1}^2 + J_{zzb} \cos \psi \dot{q}_{T1}^2 - Z_b^2 m_b \cos \psi \dot{q}_{T1}^2 - Z_b Z_h m_b \dot{q}_{T1}^2 + Z_b g m_b \cos q_{T1}) \sin \psi \end{bmatrix} \quad (27)$$

F. Rigid Rotating Blade, Tilting Nacelle code

G. Rigid Rotating Blade, Tilting Nacelle Parameters

H. Flexible Rotating Blade, Tilting Nacelle code

I. Flexible Rotating Blade, Tilting Nacelle Parameters